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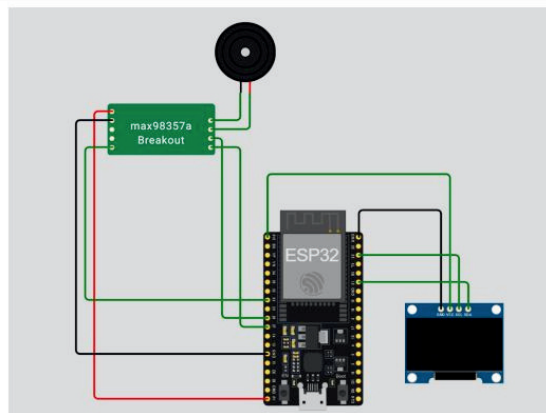
INTRODUCTION

Language practice among primary school students is often limited outside the classroom. This project introduces an IoT- and AI-based system that monitors students' spoken language during daily activities (e.g., ordering food at the canteen). Using voice recognition, OLED display, and speaker feedback, students receive real-time responses, while teachers can track performance via a database.

OBJECTIVES

- Encourage multilingual speaking in daily school life
- Detect spoken language using voice recognition
- Automatically log and store data for teacher review

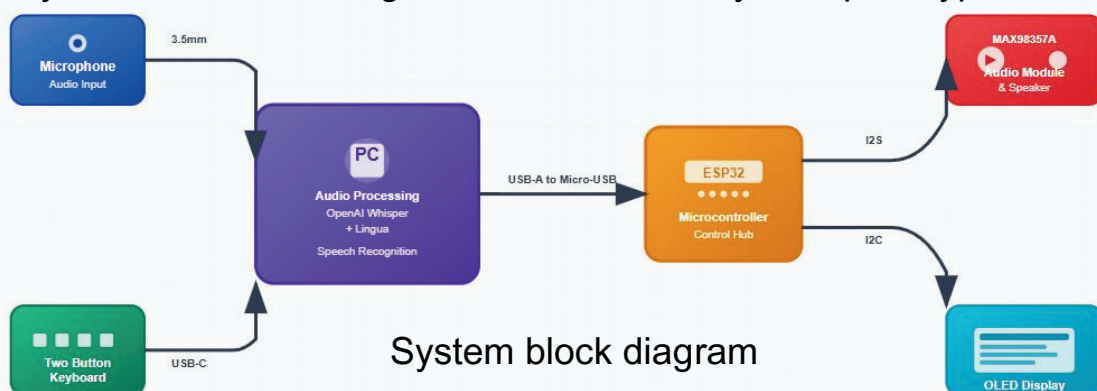
PROJECT METHODOLOGY



System schematic diagram



System prototype



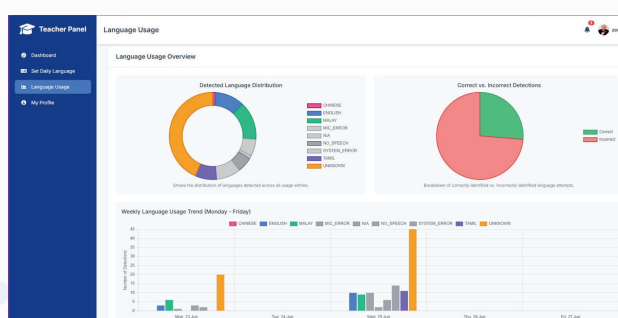
System block diagram

Project Methodology Flow

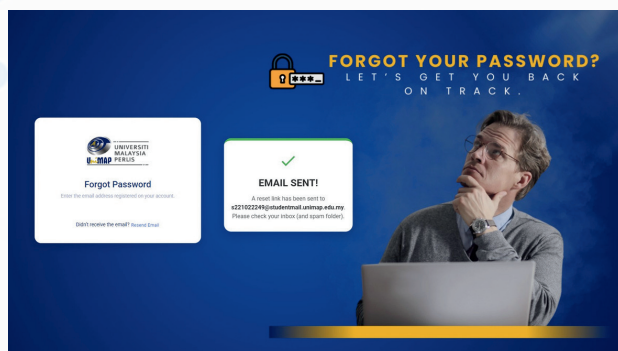


System flowchart

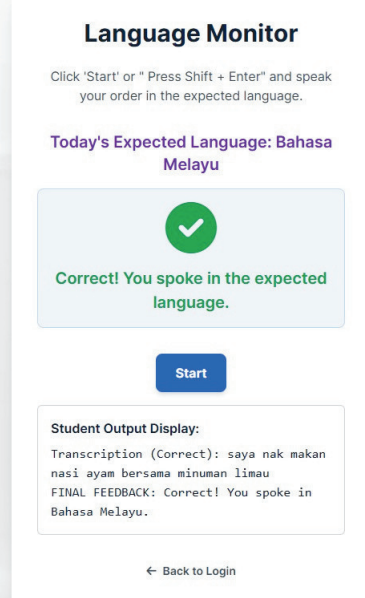
RESULT & DISCUSSION



Teacher dashboard interface



Forgot password interface



Language Monitor Interface

RELEVANCE TO SOCIETY

Cultural Impact
Inclusivity: Rotates Malay, English, Chinese, Tamil.
Multilingualism: Encourages diverse language use.
Cultural Respect: Fosters awareness and representation.

Health & Safety
School-Safe: Simple, canteen-ready design.
Privacy First: No voice data stored.
Secure Logs: Text-only, teacher-accessible.

RELEVANCE TO SDG

SDG 4 Quality Education: Supports inclusive, equitable learning via real-world language practice



Impact: Enables tracking language performance in large classes.

Sustainability: Uses low-power ESP32/OLED components.

CONCLUSION

This project successfully delivers a functional prototype that encourages multilingual speaking through daily interaction. It integrates IoT, AI, and database logging, making it suitable for real school settings. The system is affordable, scalable, and reduces teacher workload while supporting language development in young students.